Lesson Notes

MODULE 4 | LESSON 3

HOME EQUITY AS AN OPTION

Reading Time	60 minutes
Prior Knowledge	Option Types, Option Payoffs, Credit Risk, Non-Linearity
Keywords	Home equity, Long call, Short put, Recourse, Non-recourse

In the previous lesson, we outlined the features of leverage and nonlinearity. Here, we show that a mortgage represents leverage, and the home equity represents nonlinearity.

1. Benefits

Suppose you want to purchase a home, but you don't have enough funds for the entire cost. You can find a mortgage in the mortgage market. There are different types of mortgages, but there is one key property they have in common: leverage. Buying a home by financing some of the expense is exactly like buying equity with borrowed costs. In each case, you buy a valuable asset (home or stock) by borrowing money (mortgage or cash) while pledging the asset as collateral (home or stock again). Mortgages inherently have leverage through the idea of borrowed funds. Therefore, the benefits of having a mortgage apply to your house as an investment, just as there are benefits to buying stock with financing.

One difference is that home ownership is not considered speculative since you may actually live there, perhaps even as your primary residence. Some homes are secondary residences, and these tend to have different rates and terms for the mortgage. Buying and living in an 80% mortgaged home is not considered to be speculative. In particular, it is less speculative than buying stocks with 80% financing. Therefore, in real estate, the amount of leverage tends to be more generous. In U.S. markets, a regulatory agency called the SEC limits stock buyers to financing up to 50% of the stock price. However, in U.S. mortgage markets, it's possible to buy a home with just 20% down, financing 80% of the cost. Recall that if the home increases 10% in price, you would actually make $5 \times 10\% = 50\%$ on your investment.

Some people argue that homes are not straightforward investments; they are also consumables. Home ownership can be expensive in terms of the **carry cost**. The carry cost includes mortgage payments, real estate taxes, insurance costs, energy bills and other utilities, and the maintenance required due to wear and tear. Unlike financial assets, real assets have wear and tear. They tend to need storage, insurance, and maintenance--costs that are largely irrelevant for financial assets like stocks or bonds. Like financial assets, house prices are subject to volatility. It's possible to buy a home and then the price drops in value (a "housing correction"), in which case you could lose money when you sell it. Nevertheless, a house provides a real, tangible place to live.

This lesson is not meant to provide best practices on buying or financing a home but rather to illustrate one important similarity between mortgaged homes and options. To understand this similarity, let's review an idea going back to Module 1: credit risk.

One of the first models in credit risk was done by a researcher named Robert Merton. Indeed, this is the same Merton that was part of the Black-Scholes-Merton option pricing model, which earned Myron Scholes and Robert Merton the Nobel Prize in Economic Sciences in 1997. Unfortunately, Fischer Black had died prior to this date. In Merton's honor, this model is known as the Merton model.

Imagine a company that has issued stock and bonds (or more precisely, a single bond). The bond is a debt that the firm will owe at some time, T, in the future. The stock is held by the owners of the company. In his model, Merton imagined that the value of a stock can be thought of as an option. The Merton option states that the stockholders have a call option on the assets of the firm, with a strike price at the debt of the firm.

Imagine all the stock that the shareholders own. Merton said that this is equivalent to owning a call option on the assets of the firm. The strike price is the debt that the firm has. The firm is due at time T, which is the option's expiration. Let's imagine three scenarios:

1. Suppose the asset price is 120. The firm can liquidate its assets (at the market price of 120). It can repay its debt at 100. It has 20 left over to share among the stockholders. Thus, the payoff of the stockholders is positive because the assets were valuable enough to pay off the debt and have some left over.
2. Suppose the asset price is 100. The firm liquidates its assets at the market price of 100, but the entire proceeds are used to satisfy the debt. There is nothing left over for the stockholders. The stocks are worthless.
3. Suppose the asset price is 90. The firm liquidates its assets at the market price of 90, but those proceeds are insufficient to satisfy the debt. Nevertheless, the debt holders are paid the entire proceeds, which is insufficient to satisfy their debt. Once again, there is nothing left over for the stockholders.

All three scenarios have the classic formula that

$$\textbf{Equity Value} = \mathbf{max}\{\textbf{Asset}_{\textbf{Final}} - \textbf{Debt}, \mathbf{0}\}.$$

This is the exact same formula for a call option. Recognize the similar property: nonlinearity.

Equity can be positive, but it can never be negative. If a firm has insufficient assets to cover its debt, it declares bankruptcy, pays off its debtors as best it can, and writes letters to their stockholders apologizing for losing all the equity.

What does all this have to do with housing? Let's add the word "home" in front of each term.

$$\textbf{Home Equity} = \mathbf{max}\{\textbf{Home Price}_T - \textbf{Home Debt}, \mathbf{0}\}.$$

where home debt is the value of the mortgage and home equity is the homeowner's "equity" in the house. See Figure 2 for details.

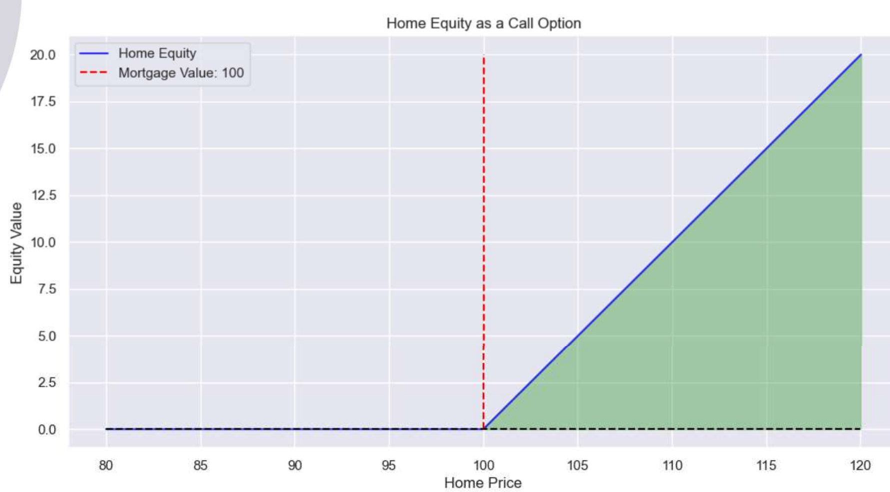
As before, the Merton model states that you have some equity ownership in the house provided the home's price exceeds the mortgage. In this case, your home equity is like a call option that is in the money. Note that the horizontal axis is the market value of the house. It's not what you paid but the current market value. Since housing prices change with market conditions, this can be more or less than your purchase price.

Let's use the same examples:

- 1. Suppose the home's market price is 120. You could sell the house at 120, repay the mortgage of 100, and pocket the difference of 20 as P&L. Thus, as an investor, you could have enjoyed a capital gain of 20 on a house.
- 1. Suppose instead the home's market price is 100. When you sell the price at 100, you use the entire proceeds to pay off the mortgage of 100. There is nothing leftover; there is no home equity. You have no capital gain.
- 1. Suppose instead the home's market price is 90. When you sell at 90, you use the proceeds to pay the mortgage, but you fall short. When a house is worth less than the mortgage on it, we say the house is "*underwater*." This is the same meaning as out of the money. Instead of paying 100, you only have 90. Will the bank want to get the other 10 from you?

Well, according to the option payoff, the answer is no. The option framework simply flattens out. The nonlinearity protects you on the downside. But is an option the appropriate model for the underwater mortgage? How is it possible to just walk away?

Figure 1: Home Equity as a Call Option



The issue has to do with whether the mortgage is *recourse* or *non-recourse*. In the United States, different states allow different types of loans: some allow non-recourse loans, and others do not. In each type of loan, the house serves as collateral for the loan. If the homeowner fails to pay the mortgage, the lender will repossess the home. If the loan is **non-recourse**, the home is the full extent of the defaulter's (homeowner's) obligation. If the loan is **recourse**, the bank can proceed to obtain other assets to pay off the mortgage in full. See the difference? The non-recourse loan is safer for the homeowner because it uses the home, and only the home, as collateral. A recourse loan is riskier for the homeowner because it means that a person can lose even more than their home: they can lose their savings and investments.

Of course, from the bank's perspective, the recourse loan is safer because they can reclaim losses when the home is worth less than the mortgage amount. Again, from the bank's perspective, the non-recourse loan is riskier because an "underwater house" is a guaranteed loss.

Which loan type represents options? The non-recourse loan, since it allows the homeowner to walk away on an underwater loan.

2. Problems

Losing a home is traumatic and financially devastating. Having further losses can intensify the pain and suffering. Some states in the U.S., therefore, have mortgages with no recourse. Other states have mortgages that allow recourse. Non-recourse loans seem less problematic.

However, suppose you combined non-recourse loans with the ability to enter the mortgage with virtually no money down. In other words, suppose you are able to buy a home with a non-recourse mortgage with 100% financing. That's right, no money down. Now you have an at-the-money option for free. You don't have to pay for this option. Why? You needed no funds down to enter and were able to borrow the strike's price. Now you have a bet:

- *If the house price increases, flip the house: sell the house, pay off the mortgage, and collect the difference.*
- *If the house price decreases, walk away.*

In the first case, you make money. In the second case, you do not lose any money. See the problem? This is an amazing deal.

It's like a free lottery ticket. There's a positive chance to make profits and virtually no chance to lose money. (Well, the reality is there are high carry costs for real estate, so there are chances to lose money.) This type of market can and will invite speculators of all kinds. Speculators will simply buy homes with little to no money down, wait and hope for the home prices to increase, and then close the trade and exit with a profit. They try to repeat this as many times as possible until the trend reverses and housing prices decline. Then, they simply walk away, leaving banks with the keys to the house.

This scenario is not hypothetical. It happened thousands and thousands of times in the U.S. during the Great Financial Crisis. Speculators flipped home after home, enjoying a real-estate bull market. Then, housing prices did not keep rising--instead they fell. Many speculators, and bona fide homeowners living in their mortgaged homes as their primary residence, defaulted on their mortgages. The homes served as collateral, which the banks then repossessed. Banks owned millions of homes that were underwater, without the ability to be made whole.

The problem of excessive speculation is that asset bubbles can be created. We will discuss this in more detail in subsequent modules.

One other note is that there is another option at play. The lender is short a put option on the assets of the firm. Let's look at their position in Figure 3.

Figure 2: Mortgage as a Short Put Option

The homeowner is long a call option on the house's value, struck at the mortgage level. The lender is short a put option on the house's value, struck at the mortgage level.

If the house price exceeds the mortgage, the proceeds will pay the lender in full. In this case, the option is out of the money. However, the lender is short, so that is good. Since they are short, they want to be out of the money.

However, if the house price is less than the mortgage, the proceeds pay the lender only in part. In this case, the option is in the money, and the lender loses some of their repayment.

In summary, giving mortgages with no money down is basically giving a free option to the homeowner. Free options are dangerous as they invite speculators and create bubbles.

3. Conclusion

In this lesson, we looked at home equity as an option on the house's value struck at the mortgage. In the next lesson, we'll review the reasons people trade options.